Forms of Inner Thinking (FIT): a prompt-based questionnaire reveals individual differences in inner language

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People vary substantially in their reported use of inner language during thought [1-4]. However, the majority of the evidence for the use of inner language comes from generic questions instead of prompting people to reflect on their experiences in the moment, which may lead to overestimating the frequency of linguistic thought [5]. Further, inner language might take different forms: *auditory* (perceived speech), *articulatory* (generated speech), *orthographic* (perceived written words), or perhaps *pure lexical* (no perceptual/articulatory form). What is the relative frequency of different forms of inner language? How much do they differ across individuals? And are they modulated by the type of inner experience (e.g., *spontaneous* thought on a given topic vs. imagining a *specific* sentence)?

To answer these questions, we developed the Forms of Inner Thinking (FIT) questionnaire. FIT instructs participants to reflect on or imagine different prompts and report the presence of language and associated sensorimotor features, nonverbal conceptual thoughts, and visual images. In addition to freely thinking about a topic (Spontaneous; 20 sec/prompt), participants also reported their experiences when imagining themselves/someone talking about a topic (General; 20 sec/prompt) or hearing/saying a sentence (Specific; 10 sec/prompt) (Fig. 1A-B). FIT was administered in English, and we report results from 227 fluent speakers. Our results focus on the Spontaneous condition, unless noted otherwise.

We found substantial variation in the forms of inner thinking. 55.5% of people always think in language and 4.4% of people never think in language [6]. In addition, 46.3% of people always think in visual images, whereas only 25.1% of people reported often or always use of nonverbal concepts (Fig. 1C). The reported form of inner language was also variable: auditory language imagery was common (45.8% of people reported experiencing it often or always), but people also reported orthographic and articulatory experiences at non-trivial frequencies (>5%; ps<.001). Moreover, 17.6% of the time, people reported experiencing words in pure lexical forms—verbal thoughts without sensorimotor features (p<.001). Auditory imagery was positively correlated with articulatory imagery (p<.001), but negatively with pure lexical (p<.001; Fig. 1D), validating the auditory/lexical distinction. These results show that perceived inner language can take different forms, sometimes lacking associated sensorimotor features.

To search for possible shared factors underlying inner thought forms, we ran an exploratory factor analysis. We identified two orthogonal latent factors: a speech imagery factor (high loadings from language, auditory, and articulatory imagery) and a pure lexical factor (high loading from pure lexical imagery) (Fig. 1E). Other imagery types were not meaningfully explained by the factor structure. This result indicates that sensorimotor aspects of spoken language rely on a shared factor, whereas pure lexical form is an additional factor that explains individual variability beyond the strength of sensorimotor imagery.

Finally, how stable are the traits observed in spontaneous thinking across different prompt types? We found that for people with low speech imagery, Specific prompts (compared to Spontaneous prompts) increased the use of language and associated sensorimotor forms (auditory, in particular, ps<.001). For the high speech imagery group, however, all forms of inner language remained stable across prompt types. Among the high lexical group, a significant modulation was similarly observed in the low speech imagery group: the use of pure lexical form decreased in Specific hearing or speaking (ps<.05). In contrast, Specific prompts consistently suppressed the frequency of nonverbal concepts and visual images (ps<.05). These results show that the stability of inner thought forms varies between people.

Altogether, using FIT we separate the use of inner language from its associated sensorimotor features, and provide evidence for the existence of a pure lexical form. Moreover,

the significant modulation by different prompt types shows that inner language imagery can be elicited even when it's not prevalent during spontaneous thought. This highlights both within and across individual variations in the use of language in thinking.

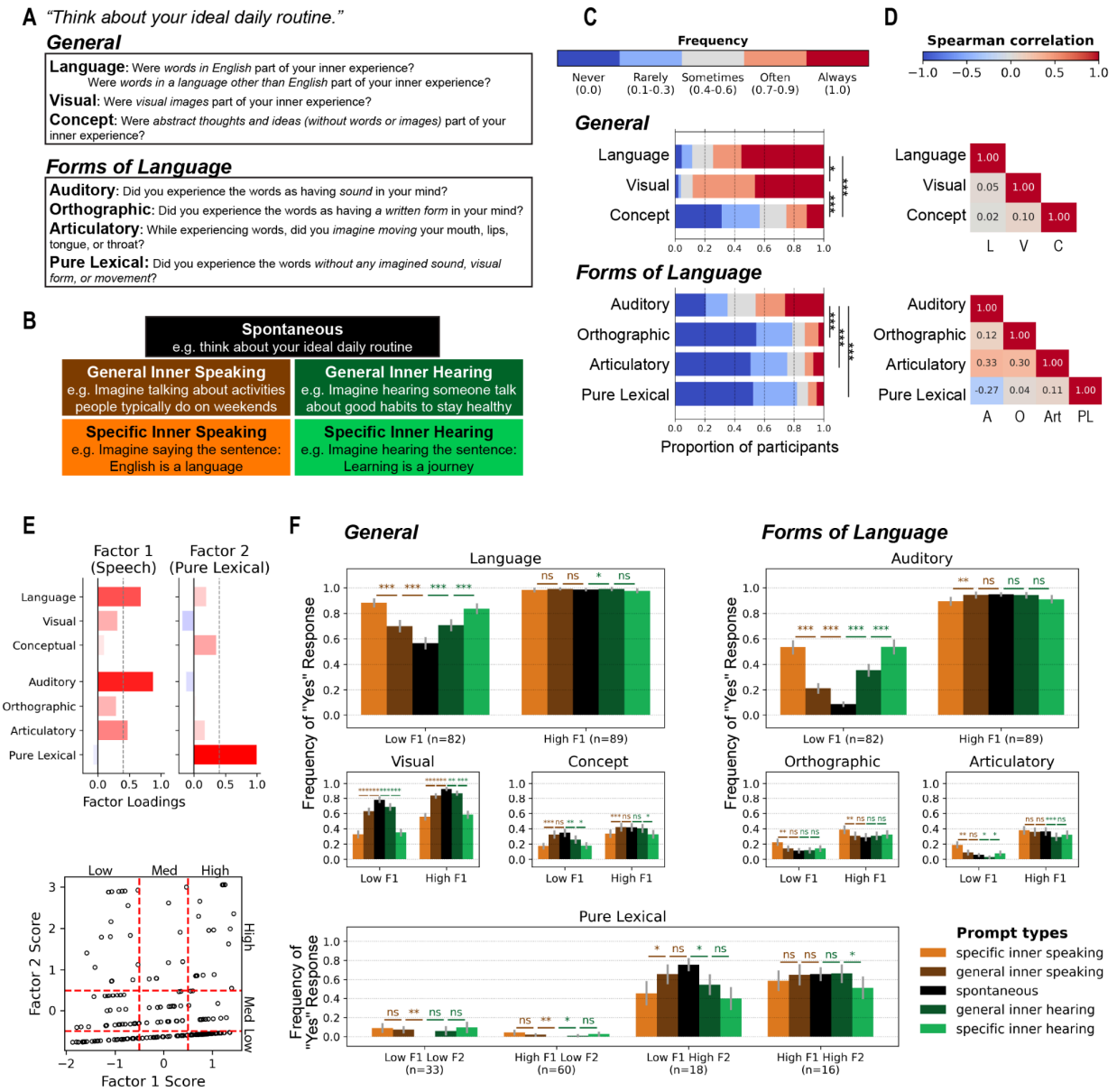

Figure 1. (A) Inner experience questions. (B) Prompt types and questionnaire item examples. (C) Distribution of how often participants answered "yes" for each question. (D) Intercorrelation between different inner experience forms. (E) Exploratory factor analysis results of inner experience forms. (F) Response profiles across prompt types by factor-score groups (high/low F1, and high/low F1 × high/low F2 for pure lexical). The *p* values are corrected using the Holm-Bonferroni method for multiple comparisons. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (corrected).

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